5. 配置(Config)：示例附页

返回配置译文与科普图。以下保留原文的 Python、YAML 和 CLI 示例；它们涉及不同模型，用于说明配置写法。

声明阶段(Declarative Config)

这是多模态流水线的配置片段，省略了部分阶段定义；点分路径中的 ... 也是占位符，不能直接作为完整配置运行。

```
# 非 TP 阶段必须显式声明 process，没有隐式默认值。
# 此处每个阶段独占一个 OS 进程；设置相同 process 可以共用进程。
stages = [
  StageConfig(
    name="preprocessing",
    process="preprocessing",
    factory_path="...create_preprocessing_executor",
    next=["image_encoder", "audio_encoder", "mm_aggregate"],
    project_payload={
      "image_encoder": "...project_preprocessing_to_image_encoder",
      "audio_encoder": "...project_preprocessing_to_audio_encoder",
      "mm_aggregate": "...project_preprocessing_to_mm_aggregate",
    },
  ),
  StageConfig(
    name="mm_aggregate",
    process="mm_aggregate",
    factory_path="...create_aggregate_executor",
    wait_for=["preprocessing", "image_encoder", "audio_encoder"],
    merge_fn="...merge_for_thinker",
    next="thinker",
  ),
  EngineStageConfig(  # 使用 SGLang 引擎，因此支持 engine.*
    name="thinker",
    process="thinker",
    factory_path="...create_sglang_thinker_executor_from_config",
    factory=FactoryArgs(max_seq_len=8192),
    gpu=0,
    next=["decode", "talker_ar"],
    stream_to=["talker_ar"],
  ),
  StageConfig(
    name="decode",
    process="decode",
    factory_path="...create_decode_executor",
    terminal=True,
  ),
]
```

工厂专属参数(Factory-specific Settings)

```
stages:
 latent_engine:
  factory:
   num_steps: 4  # FactoryArgs 未声明；按 num_steps=4 传给工厂，由签名验证
```

YAML 配置

```
config_cls: MossTTSPipelineConfig
model_path: OpenMOSS-Team/MOSS-TTS

stages:
 tts_engine:
   tp_size: 2
   engine:
     mem_fraction_static: 0.7
 vocoder:
   factory:
     dtype: bfloat16
```

命令行覆盖(CLI Overrides)

```
sgl-omni serve --config omni.yaml \
   --tts_engine.tp_size 2 \
   --tts_engine.engine.mem_fraction_static 0.7 \
   --vocoder.factory.dtype bfloat16 \
   --vocoder.process vocoder
```

共享配置(Shared Values)

```
shared:
 - select:
     stages: [preprocessing, latent_engine]  # 也可用 engine: true、exclude: [...]
   factory:
     num_steps: 4
```

工厂参数通道(Factory Argument Channels)

```
PipelineConfig.stage_factory_kwargs(name) ← 作者通道：代码中的组件连接
stage.factory.*                  ← 配置通道：按字段名传入
stage.engine.* → server_args_overrides    ← 配置通道：一个字典传给 SGLang
```

运行时文件(Runtime Files)

```
pipeline/
 ├── stage_workers.py  ← StageLaunchConfig、子进程入口、StageGroup
 ├── runtime_config.py ← 端点、运行时目录与设备放置准备
 └── mp_runner.py      ← 跨阶段编排、管理协调器(Coordinator)
```